

Field and Galois Theory

Food and Shelter

1. Give an example of a field F and an algebraic extension of F which has infinite degree.
2. (**August 2017 Problem 4**) Suppose that $K \subseteq \mathbb{C}$ is a Galois extension of $\mathbb{Q}$, $[K : \mathbb{Q}] = 4$, and $\sqrt{-m} \in K$ for some positive integer m . Show that $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Self-Actualization

3. (January 2019 Problem 4)

- (a) Let E be the splitting field over $\mathbb{Q}$ of $e(x) = x^3 + 2x^2 + 3x + 4$, which has discriminant -200 . Find $\text{Gal}(E/\mathbb{Q})$.
- (b) Let F be the splitting field over $\mathbb{Q}$ of $f(x) = x^3 + 4x^2 + 7x + 6$, which has discriminant -72 . Find $\text{Gal}(F/\mathbb{Q})$.
- (c) Find $\text{Gal}(EF/\mathbb{Q})$.

4. (January 2018 Problem 5)

Let K be the splitting field of the polynomial $f(x) = x^4 - x^2 - 1$ over $\mathbb{Q}$. Compute $\text{Gal}(K/\mathbb{Q})$. (For partial credit, find the degree $[K : \mathbb{Q}]$.)

5. (August 2022 Problem 4)

Let p be a prime number, $\mathbb{F}_p$ the finite field with p elements, and $K = \mathbb{F}_p(t)$ the field of rational functions in one variable t over $\mathbb{F}_p$. (So that K is the fraction field of the polynomial ring $\mathbb{F}_p[t]$.)

- (a) Show that the splitting field of the polynomial $x^p - t \in K[x]$ over K is inseparable.
- (b) Show that the splitting field of $x^p - t - 1 \in K[x]$ over K is the same as the splitting field of $x^p - t$.
- (c) Show that K has a single degree p inseparable extension.

6. (January 2022 Problem 5)

Let F be a field of prime characteristic p . Suppose $E = F(\alpha)$ is a simple extension such that $\alpha \notin F$ but $\alpha^p - \alpha \in F$.

- (a) Find $[E : F]$.
- (b) Prove that E/F is a Galois extension.
- (c) Find the Galois group $\text{Gal}(E/F)$.

Hint: What is $(x + 1)^p - (x + 1)$ in characteristic p ?

7. (August 2015 Problem 5) Let $\mathbb{C}(x)$ be the field of rational functions with complex coefficients in the variable x . Thus, x is transcendental over $\mathbb{C}$. Put

$$y = x^n + x^{-n} \in \mathbb{C}(x)$$

for some $n > 0$. Prove that the field extension $\mathbb{C}(x)/\mathbb{C}(y)$ is a finite Galois extension and find its degree.

8. (August 2020 Problem 3) Let L/K be a degree p Galois extension, where p is prime, and let σ be a nontrivial element of $\text{Gal}(L/K)$. Then σ is a linear transformation of the K -vector space L .

- (a) What is the characteristic polynomial of σ ?
- (b) Write L^0 for the subspace of L consisting of elements with trace 0. Show that σ sends L^0 to L^0 .
- (c) What is the characteristic polynomial of σ acting on L^0 ?

9. (January 2022 Problem 2) For each of the following statements, either prove it or provide a counterexample.

- (a) Every matrix of finite order in $\text{GL}_n(\mathbb{C})$ is diagonalizable.
- (b) Every matrix of finite order in $\text{GL}_n(\mathbb{Q})$ is diagonalizable.
- (c) Every matrix in $\text{GL}_n(\overline{\mathbb{F}}_p)$ has finite order. (We write $\overline{\mathbb{F}}_p$ for the algebraic closure of the finite field $\mathbb{F}_p$.)
- (d) Every matrix of finite order in $\text{GL}_n(\overline{\mathbb{F}}_p)$ is diagonalizable.